

1. $\pi_{SNO, SName, Status, City}(Supplier)$
2. $\pi_{PNO, PName, Color, Weight, City}(Part)$
3. $\sigma_{City='London'}(Supplier)$
4. $\pi_{PNO, PName, Color}(\sigma_{City='Paris'}(Part))$
5. $\pi_{PNO, PName}(\sigma_{Weight>15}(Part))$
6. $\pi_{PNO, PName}(\sigma_{Weight>15 \wedge Color \neq 'red'}(Part))$
7. $\pi_{PNO, PName}(\sigma_{Weight>15 \wedge Color \neq 'red' \wedge Color \neq 'green'}(Part))$
8. $\pi_{PNO, PName}(\sigma_{Weight \geq 15 \wedge Weight \leq 19}(Part))$
9. $\pi_{PNO, PName, Color, Weight, City}(\sigma_{Shipment.SNO='S1'}(Shipment \bowtie_{Shipment.PNO=Part.PNO} Part))$
10. $\pi_{SNO, SName, Status, City}(\sigma_{Shipment.PNO='P1'}(Shipment \bowtie_{Shipment.SNO=Supplier.SNO} Supplier))$
11. $\pi_{SNO, SName, Status, City}(\sigma_{Supplier.City='London' \wedge Part.City='London'}((Supplier \bowtie_{Supplier.SNO=Shipment.SNO} Shipment) \bowtie_{Shipment.PNO=Part.PNO} Part))$
18. $T1 \leftarrow \pi_{SNO, SName, Status, City}(Supplier \bowtie_{Supplier.SNO=Shipment.SNO} Shipment)$
 $T2 \leftarrow Supplier$
 $T3 \leftarrow T2 - T1$
20. $T1 \leftarrow Supplier \bowtie_{Supplier.SNO=Shipment.SNO} Shipment$
 $T2 \leftarrow \sigma_{Shipment.SNO \text{ IS NULL}}(T1)$
21. $\gamma_{COUNT(SNO)} \rightarrow TotalSuppliers(Supplier)$
22. $\gamma_{COUNT(SNO)} \rightarrow TotalSuppliers(\sigma_{City='London'}(Supplier))$
23. $\gamma_{MAX(Status)} \rightarrow MaxStatus, MIN(Status) \rightarrow MinStatus(Supplier)$
24. $\gamma_{MAX(Status)} \rightarrow MaxStatus, MIN(Status) \rightarrow MinStatus(\sigma_{City='London'}(Supplier))$

25. $\gamma_{SNO;SUM(Qty) \rightarrow TotalShipped}(Shipment)$
26. $T1 \leftarrow \gamma_{SNO;SUM(Qty) \rightarrow TotalShipped}(Shipment)$
 $T2 \leftarrow \pi_{Supplier.SNO, SName, City, TotalShipped}(Supplier \bowtie_{Supplier.SNO=T1.SNO} T1)$
27. $\pi_{SNO}(\sigma_{TotalShipped > 500}(\gamma_{SNO;SUM(Qty) \rightarrow TotalShipped}(Shipment)))$
28. $T1 \leftarrow \sigma_{Color='red'}(Shipment \bowtie_{Shipment.PNO=Part.PNO} Part)$
 $T2 \leftarrow \gamma_{SNO;SUM(Qty) \rightarrow RedTotal}(T1)$
 $T3 \leftarrow \pi_{SNO}(\sigma_{RedTotal > 300}(T2))$
29. $T1 \leftarrow \sigma_{Color='red'}(Shipment \bowtie_{Shipment.PNO=Part.PNO} Part)$
 $T2 \leftarrow \gamma_{SNO;SUM(Qty) \rightarrow RedTotal}(T1)$
 $T3 \leftarrow \pi_{Supplier.SNO, SName, City, RedTotal}(Supplier \bowtie_{Supplier.SNO=T2.SNO} T2)$
30. $\gamma_{City;COUNT(SNO) \rightarrow NumSuppliers}(Supplier)$
31. $T1 \leftarrow \gamma_{SNO;SUM(Qty) \rightarrow TotalShipped}(Shipment)$
 $T2 \leftarrow \gamma_{;MAX(TotalShipped) \rightarrow MaxShipped}(T1)$
 $T3 \leftarrow \sigma_{T1.TotalShipped=T2.MaxShipped}(T1 \times T2)$
 $T4 \leftarrow \pi_{SName, Supplier.SNO}(Supplier \bowtie_{Supplier.SNO=T3.SNO} T3)$
32. $\pi_{City}(Supplier) \cup \pi_{City}(Part)$